Macromolecular Research

Article

DOI 10.1007/s13233-019-7036-1

www.springer.com/13233 pISSN 1598-5032 eISSN 2092-7673

Enhanced Olefin and CO_2 Permeance Through Mesopore-Confined Ionic Liquid Membrane

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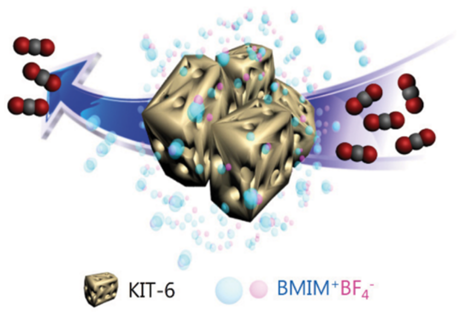
Received April 28, 2018 / Revised August 8, 2018 / Accepted August 18, 2018

Abstract: Nanocomposites of ionic liquid, 1-butyl-3-methylimidazolium tetrafluoroborate ($BmimBF_4$), in three-dimensional mesoporous silica (KIT-6) were fabricated and utilized as mesopore-confined ionic liquid membranes for Olefin and CO_2 Separation. Compared to neat $BmimBF_4$, the fabricated membrane showed 10-times-enhanced gas permeation property for olefin separation, and 5-times-enhanced gas permeation property for CO_2 separation with gas-selectivity remaining unchanged. Moreover, the enhanced diffusion-limited current density of the electrochemical cell involving I^-/I_3^- redox couple in $BmimBF_4$/KIT-6 nanocomposite was observed. This implies that mesoporous KIT-6 can be used as an effective additive in nanocomposite membranes for enhanced olefin and CO_2 permeance.



Keywords: mesoporous materials, ionic liquids, CO_2, olefin separation, and iodine diffusion.

1. Introduction

Highly ordered mesoporous materials have attracted great interest because of their unique nanostructure with high specific surface area and pore volume.[1-3] In particular, three-dimensional (3D) interconnected mesopores have continued to receive tremendous attention due to their nature and access of the pore system. Such mesoporous materials are consider to be highly advantageous for mass diffusion and transportation compared to other mesoporous structures with one- or two-dimensional channels.[4] However, recent several experimental studies of certain ionic liquids (ILs) under confinement in mesopores have demonstrated a decrease in the diffusivity due to the pore wall-IL interaction.[5-8] Conversely, Shi et al. predicted an increase of self-diffusion coefficients in ILs confined in carbon nanotubes through molecular dynamics and Monte-Carlo simulations.[9] Also, Iacob *et al.* presented the experimental evidence of 2 orders-enhancement of molecular mobility and ionic conductivity of a confined 1-butyl-3-methylimidazolium tetrafluoroborate ($BmimBF_4$) under a mean 7.5 nm pore diameter.[10] Herein, we present the first results of the application of 3D interconnected mesoporous silica (KIT-6) filled with $BmimBF_4$ for fabricating highly permeable gas-separation membranes.

For developing high-performance membrane-based gas-separation systems, we recently proposed a carrier-mediated mass transport, *i.e.*, facilitated transport for the separation of olefin/paraffin or CO_2/N_2 mixture.[11-13] In such membranes comprising specific compounds as the gas carriers in polymer matrix or ILs, the permeation performance of the target molecules increases through a carrier-mediated mass transport; consequently, a high selectivity can be obtained. For instance, 1-butyl-3-methylimidazolium tetrafluoroborate ($BmimBF_4$)/Ag nanocomposite membrane is reported to demonstrate a propylene/propane selectivity of 17 and a total mixed-gas permeance of 2.7 GPU (GPU=gas permeation unit, 1 GPU=1×10^{-6} cm^3 (STP)/(cm^2 s cm Hg); STP=standard temperature and pressure).[11] We also reported that the olefin-separation performance by neat IL media was mainly due to specific and reversible interactions of ILs such as $BmimBF_4$ and 1-methyl-3-octylimidazolium nitrate ($MoimNO_3$) with propylene.[12] Furthermore, imidazolium-based room-temperature ILs have been widely investigated in the separation of not only olefin/paraffin mixture gases but also acidic gases such as SO_2, H_2S, and CO_2 from CH_4 and N_2.[14-23] However, such IL system showed relatively low permeance, which is usually even more challenging than selectivity for industrial applications.

Taking the advantage of both ILs and KIT-6 toward mass-transport phenomena, in this study, we expected that a faster penetrant diffusion should occur in a $BmimBF_4$/KIT-6 nanocomposite through the confinement effects and a systematic manipulation of the diffusion path. Unusually, *ca.* 10-times-enhanced gas permeation property was demonstrated, with the selectivity remaining unchanged. The enhanced diffusion property of the solute in the media (ILs) was also supported by the diffu-

Acknowledgments: This work was supported by an Energy Efficiency & Resources of the Korea Institute of Energy Technology Evaluation and Planning (KETEP) (No. 20122010100040) Y. S. Kang also acknowledges the Basic Science Research Program through the NRF of Korea grant funded by the MSIP for the Center for Next Generation Dye-sensitized Solar Cells (No. 2011-0001055). S. W. Kang also acknowledges the Basic Science Research Program (2017R1D1A1B03032583) through the National Research Foundation of Korea (NRF), funded by the Ministry of Science, ICT, and Future Planning.

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sion-limited current density using an electrochemical cell involving I^-/I_3^- redox couple. Thus, the application of $BmimBF_4$/KIT-6 nanocomposite for the accelerated facilitated transport in olefin/paraffin-, CO_2/CH_4-, and CO_2/N_2-separation membranes was investigated.

2. Experimental

2.1. Preparation of KIT-6

The preparation of mesoporous silica KIT-6 was based on a report of Ryoo's group.[24] Pluronic P123 ($EO_{20}PO_{70}EO_{20}$, MW=5800, as structure-directing agent), tetraethoxysilane (TEOS, as a silica source), HCl, and all the chemicals for preparing KIT-6 were used as received from Aldrich Chemical Co.

2.2. Nanocomposites $BmimBF_4$/KIT-6 membrane

KIT-6 was added to 1-butyl-3-methylimidazolium tetrafluoroborate ($BmimBF_4$, obtained from C-TRI, Korea). After stirring the solution of IL and KIT-6 composites for 24 h, the $BmimBF_4$/KIT-6 nanocomposite coated onto a polysulfone microporous membrane support (Woongjin Chemical Industries Inc., Seoul, Korea) using an RK Control Coater (Model 101, Control Coater RK Print-Coat Instruments Ltd., UK). The selective layer was estimated to be *ca.* 20 μm using the energy dispersive X-ray (EDX) attached to scanning electron microscopy (SEM) in Supplementary material. The average pore size of the surface of the macroporous membrane support was 0.1 μm, whereas the thickness of the asymmetric structure was ~40 μm. Ionic liquids, and fumed silica powder (7 nm) was from Aldrich Chemical Co.

2.3. Membrane performances

The $BmimBF_4$/KIT-6 membrane was directly equipped with a permeation cell. The flow rates of the mixed and sweep gases (helium) were controlled using mass-flow controllers. The all-gas flow rates represented by GPU were determined using a mass flow meter in its steady state. The gas flow rates and gas permeances were measured using a mass flow meter at an upstream pressure of 40 psig and at atmospheric pressure downstream. The 50:50 propylene/propane mixed gas-separation properties of the $BmimBF_4$/KIT-6 membranes with a membrane area of 2.25 cm^2 were evaluated using a gas chromatograph (Hewlett-Packard G1530A, MA) equipped with a thermal conductivity detector (TCD) and a unibead 2S 60/80 packed column.

The gas permeances of these membranes for CO_2, N_2, and CH_4 were measured using a bubble meter. The single-gas permeance (**Q**) was measured for CH_4, N_2 and CO_2 at a pressure difference of 2 atm, representing the pressure-normalized mass flux is mathematically expressed as

$$J=\mathbf{Q}\times\Delta p$$

where J is the mass flux, and Δp is the pressure difference across the membrane. The permeance was determined from the volume of permeate gas. The selectivity of the gas permeance is defined as the ratio of the CO_2/N_2 and CO_2/CH_4 permeance.

2.4. Preparation of the electrochemical cell

For investigation of diffusion-limited current density of $BmimBF_4$/KIT-6 medium, an electrochemical cell prepared according to Houch's report.[25] Symmetrical dummy cells were assembled using two identical Pt-electrodes with the same composition used to fabricate the DSCs. Pt electrode was prepared by spreading a drop of H_2PtCl_6 solution (2 mg Pt in 1 mL ethanol) onto an FTO glass substrate followed by heating at 400 °C for 15 min. The $BmimBF_4$/KIT-6 electrolyte consisted of 0.6 M 1-butyl-3-methylimidazolium iodide (BMII, Solaronix), 0.05 M iodine (I_2) (Sigma-Aldrich, ACS reagent, 99%).

2.5. Charaterizations

Transmission electron microscopy (TEM) images were obtained by JEOL JEM-3000, operating at 300 kV. Samples for TEM imaging were prepared from a copper grid where the solution was dropped. The FT-Raman spectra were collected for the CuNPs/ILs/GO films at room temperature using JASCO NRS-3100 at a resolution of 1.0 cm^{-1}. Cyclic voltammetry (CV; Autolab potentiostat/galvanostat) was performed in symmetrical dummy cells.

Nitrogen adsorption-desorption measurements of porous silicas were performed on an Autosorb 1 instrument (Quantachrome Instruments) at 77 K. Samples were preheated at 120 °C for 3 h under 1×10^{-2} Torr prior to the measurements. The pore size distributions and specific surface areas were evaluated by Barrett-Joyner-Halenda (BJH) and Brunauer-Emmett-Teller (BET) method, respectively. Also, density functional theory (DFT) method using non-local density functional theory (NLDFT) adsorption branch model were used for the textural parameters of KIT-6.

3. Results and discussion

The highly ordered mesostructures of KIT-6 is confirmed by TEM image in Figure 1(a). N_2 adsorption/desorption isotherm is of type IV, which is typical for mesoporous materials in Fig-

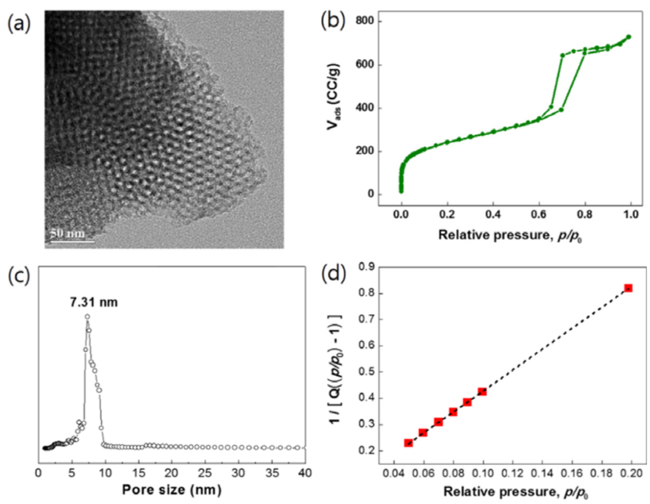


Figure 1. (a) The TEM image of KIT-6. (b) N_2 adsorption-desorption isotherms and corresponding (c) DFT pore size distribution curve, and (d) BET-plot for KIT-6.

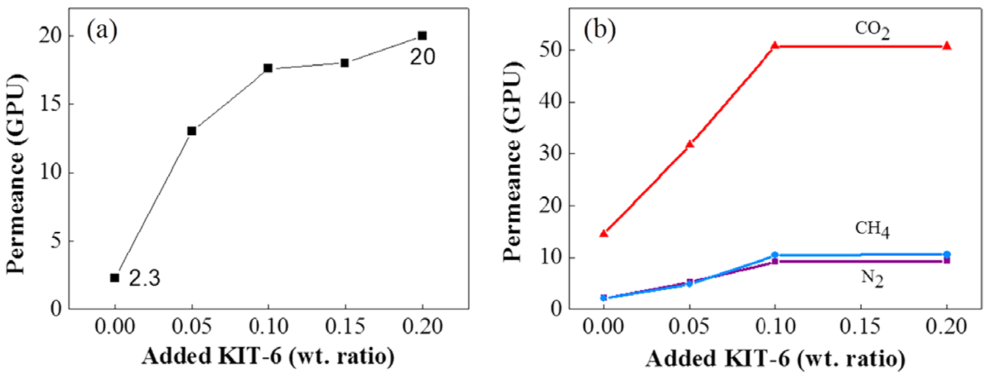


Figure 2. Permeance of $BmimBF_4$/KIT-6 membranes for separation of (a) propylene/propane (50:50 v/v) mixed gas with the mixed-gas selectivity remained constant at 3.2, and (b) CO_2 from CH_4 and N_2.

ure 2(b). The pore diameter calculated by the BJH method is 6.62 nm; On the other hand, the DFT model assuming a cylindrical pore geometry gives diameters of 7.31 nm for the main pores and specific surface area of 754 m^2/g. Terasaki *et al.* reported that the DFT model was in good agreement with the 3D reconstructed structure, which was consisted by two parts, *i.e.*, main pore and complementary pores.[26] The BET surface area is 866 m^2/g, which was calculated by the multipoint BET model from the adsorption data, as shown Figure 1(d). As expected, the high structural order of the large-pore cubic Ia3d KIT-6 materials with a narrow mesopore (5~10 nm) distribution was prepared, which are in good agreement with those reported previously for similar samples.[24]

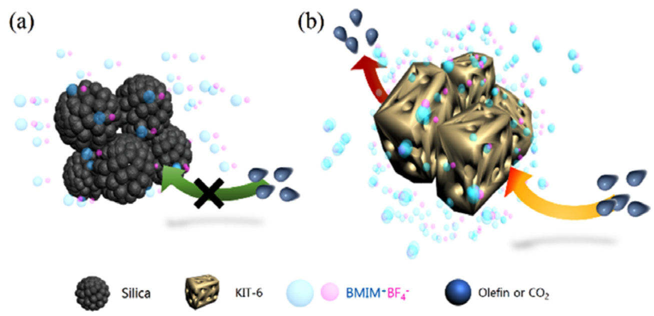
The propylene/propane (50:50 v/v) mixture-separation performances of $BmimBF_4$/KIT-6 membranes with various composition of KIT-6 were evaluated. Figure 2(a) shows that the neat $BmimBF_4$ membrane (*i.e.*, $BmimBF_4$ membrane without KIT-6) exhibited a relatively low permeance performance: The gas permeance was 2.3. Surprisingly, the introduction of 0.05 weight ratio of KIT-6 to the $BmimBF_4$ membrane increased the permeance from 2.3 to 13.4, whereas the mixed-gas selectivity remained constant at 3.2 in our previous report. Ultimately, the maximum permeance 20 was obtained in $BmimBF_4$/KIT-6 (1:0.2 w/w). In parallel, for CO_2 separation, $BmimBF_4$/KIT-6 (1:0.1 w/w) increased the permeance from 17 to 51; the other gases such as CH_4 and N_2 also increased the permeance from 2.14 to 9.33 and 10.58, respectively. Ideal separation factor and CO_2 permeances are summarized in Table 1.

There have been several challenge to increase the diffusivity of ionic liquid for CO_2 separation. At the similar condition, *i.e.*, liquid membranes using $BmimBF_4$ on support, for example, Zhao *et al.* reported that a small addition of water in $BmimBF_4$ decreased viscosity of IL, which led to increase CO_2 permeance from 11.5 to 13.8 GPU with CO_2/N_2 selectivity of 60.[27] Here, after the addition of KIT-6 into $BmimBF_4$, notwithstanding the change from liquid-state to quasi-solid-state, the increased diffusivity of the media regarding gas penetration were observed.

Interestingly, the highly enhanced permeance of $BmimBF_4$/KIT-6 nanocomposite was observed to be better than that of neat $BmimBF_4$ membrane for the same thickness of selective layer and almost same selectivity performance. We also examined the effect of a mesoporous network of fumed silica in $BmimBF_4$ for investigating the gas permeance behavior. However, $BmimBF_4$/

Table 1. Selectivity and CO_2 permeance of neat $BmimBF_4$ and various $BmimBF_4$/KIT-6 composite membranes under 2 atm pressure difference at room-temperature

$BmimBF_4$/KIT-6 composition (w/w)	Selectivity (CO_2/N_2)	Selectivity (CO_2/CH_4)	CO_2 permeance (GPU)
No KIT-6	5.0	4.8	17.0
1:0.05	6.0	6.6	31.7
1:0.10	5.1	4.2	51.7
1:0.20	5.4	4.8	51.6



Scheme 1. Fumed silica (a) and KIT-6 (b) in $BmimBF_4$ for olefin or CO_2 separation.

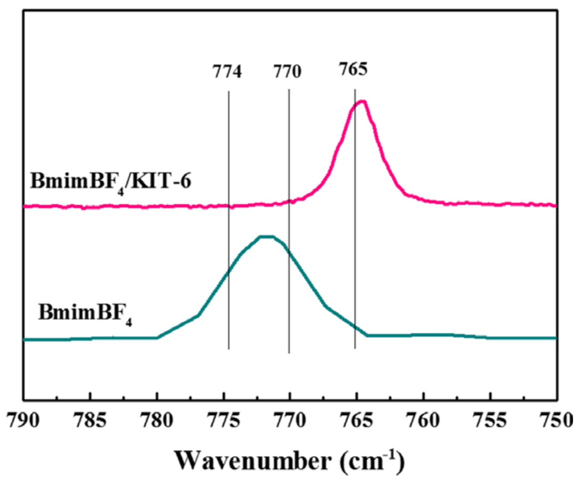


Figure 3. FT-Raman spectra: $BmimBF_4$/KIT-6 gelled powder composite and neat $BmimBF_4$ in the BF_4^- stretching region.

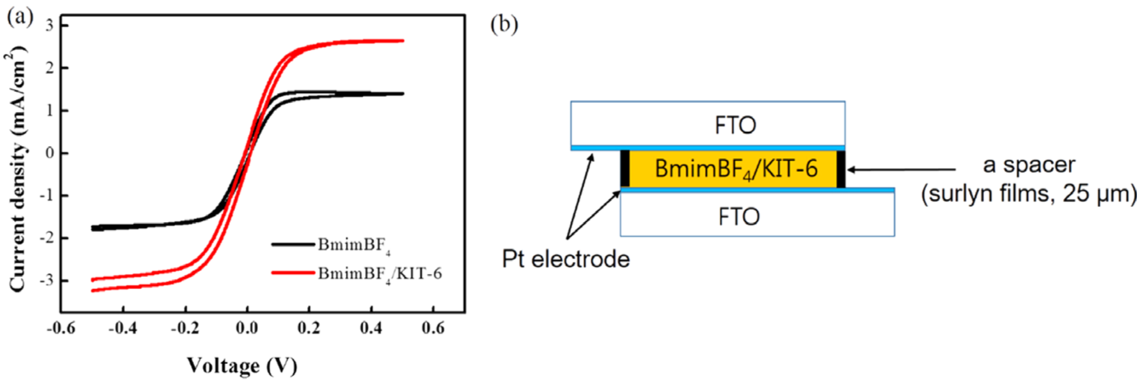


Figure 4. (a) Steady-state current-voltage curve of MPII-based liquid and BmimBF$_4$/KIT-6 (1:0.05 w/w) gel electrolytes; scan rate: 5 mV/s, and (b) The electrochemical cell used for the measurements.

silica (1:0.05 w/w) showed a slow gas transport that could not be measured. Similarly, Le Bideau *et al.* reported a decrease in the diffusivity of ILs in monolithic silica matrices.[28] Thus, the mesoporous network of fumed silica *without mesopores* may act as a barrier of gas penetration, as shown in Scheme 1.

The interactions between BmimBF$_4$ and KIT-6 in 1/0.1 BmimBF$_4$/KIT-6 gelled powder composites were investigated by Fourier transform (FT)-Raman spectroscopy. The Raman spectra in the regions of BF_4^- stretching bands are shown in Figure 3. The peaks for free ions, ion pairs, and higher-order ion aggregates of BF_4^- appeared at 765, 770, and 774 cm^{-1},[11] respectively. Only the peak intensity at 765 cm^{-1}, corresponding to free BF_4^- anions, was observed in BmimBF$_4$/KIT-6 (1:0.1 w/w). In contrast, in the absence of KIT-6, neat BmimBF$_4$ showed the almost aggregated and ion-paired states of anions. This increase can be explained by the fact that the introduction of inorganic nanoparticles greatly increases the concentration of free anions. Therefore, the use of KIT-6 as the gelator in BmimBF$_4$ was successfully demonstrated by the favored interactions between the KIT-6 and BF_4^-.

Finally, the diffusion media property of BmimBF$_4$/KIT-6 was evaluated using an electrochemical cell, which was designed based on the report on DSC application as shown in Figure 4 (b).[25] The cell was filled with the electrolyte: 0.6 M of 1-methyl-3-propylimidazolium iodide (MPII) and 0.05 M of iodine (I_2) in BmimBF$_4$/KIT-6 (1:0.05 w/w) nanocomposite.

Figure 4 shows the comparative steady-state voltammograms for BmimBF$_4$/KIT-6 obtained at the scan rate of 5 mV/s. The diffusion constant of I_3^- was calculated from the diffusion-limited current density (j_{lim}) by using Eq. (1) (Table 2).

$$D_{I_3^-} = \frac{l}{2nFc_{I_3^-}} j_{lim} \tag{1}$$

where n is the electron number per molecule, F is the Faraday constant, and $c_{I_3^-}$ is the concentration of I_3^- species. Because of large excess of I^- in the system, only the diffusion of I_3^- limits the current. As shown in Table 1, the diffusion constant of I_3^- ($D_{I_3^-}$) in BmimBF$_4$ increased from 1.84 to 3.75 when 0.05 weight ratio of KIT-6 was added, which is 87.5% of the increased value and is consistent with the gas-separation performances of the membranes. Therefore, this result clearly showed that by utilizing 3D interconnected mesopores such as KIT-6, hybridization can be used as one of the effective strategies for enhancing the mass-transport properties.

Table 2. Diffusion-limited current (j_{lim}) and diffusion constant ($D_{I_3^-}$) of I_3^- in BmimBF$_4$ in the absence of KIT-6.

	j_{lim} (mA/cm^2)	$D_{I_3^-}$ (cm^2/s)
BmimBF$_4$	1.42	1.84×10^{-7}
BmimBF$_4$/KIT-6	2.66	3.45×10^{-7}

4. Conclusions

In summary, KIT-6 was used as an additive to maximize the permeance performance of highly selective membranes for gas separation. Such increased diffusion properties are mainly attributed to the nano-confined ionic liquid effects and a systematic manipulation of the diffusion path. We believe that such a quasi-solid-state electrolyte offers specific benefits over the ILs for gas separation and provides the new insights in dynamics of IL under confinement.

Supporting information: Information is available regarding the cross section of the BmimBF4/KIT-6 (1:0.1 w/w) membrane. The materials are available *via* the Internet at http://www.springer.com/13233.

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